ITAI-1371-Intro to Machine Learning-RT-14274

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Lab 10 Unsupervised Learning

Reflective Journal

This lab helped our team understand how unsupervised learning differs from the supervised learning methods we had been using in previous labs. Before this lab, most of our machine learning work involved labeled data, where we had features and a known target variable. In this exercise, we worked without a target and learned how a model can discover patterns and structure directly from the data.

The first major concept we explored was K-Means clustering. We used a synthetic dataset representing customer spending patterns and applied the Elbow Method to determine an appropriate number of clusters. This helped us understand that clustering is not about predicting a known answer. Instead, the goal is to find natural groupings among similar data points.

The Elbow Method was especially useful because it showed us that choosing more clusters does not always produce a more meaningful solution. As the number of clusters increases, inertia decreases, but eventually the improvement becomes much smaller. In our results, the elbow appeared around k = 4, which matched the four visible groups in the synthetic data. This helped us understand the idea of balancing model complexity with useful interpretation.

After selecting four clusters, we applied K-Means and visualized the results. Seeing the points separated into different groups and the cluster centers marked on the graph made the algorithm easier to understand. We could see that K-Means assigns each observation to the nearest centroid and then adjusts the centroids until the groups stabilize. This visual result made clustering feel more intuitive than simply reading the mathematical description.

The second major concept in the lab was Principal Component Analysis, or PCA. PCA introduced us to dimensionality reduction, which was different from clustering because the purpose was not to create groups. Instead, PCA transformed the original features into a smaller number of new components while trying to preserve as much information as possible.

We applied PCA to the Iris dataset, which originally contained four features, and reduced it to two principal components. The first two components explained approximately 97.77% of the total variance. This result was important because it showed that most of the meaningful information in the original four-dimensional dataset could be represented using only two dimensions.

This helped us understand why dimensionality reduction can be useful in machine learning. Working with fewer dimensions can make data easier to visualize and can also reduce the complexity of future models. At the same time, we learned that reducing dimensions involves a tradeoff because some information can be lost. In this case, the loss was relatively small because the first two components preserved almost all of the variance.

One of the most important lessons from this lab was that unsupervised learning requires a different way of thinking. In supervised learning, we can compare predictions with known correct answers. In unsupervised learning, there may not be one exact “correct” result. We instead evaluate whether the patterns or groups we discover are meaningful and useful for the problem we are trying to solve.

The lab also helped us connect these techniques to real-world applications. K-Means could be used for customer segmentation, grouping products, or identifying similar behaviors. PCA could be useful when a dataset contains many features and we want to simplify it before visualization or further analysis.

Overall, this lab expanded our understanding of machine learning beyond prediction. We learned that machine learning can also be used to explore data, discover hidden structures, and simplify complex datasets. K-Means showed us how clustering can reveal natural groups, while PCA demonstrated how dimensionality reduction can preserve important information using fewer variables. These concepts helped us see unsupervised learning as an important part of the larger machine learning workflow.